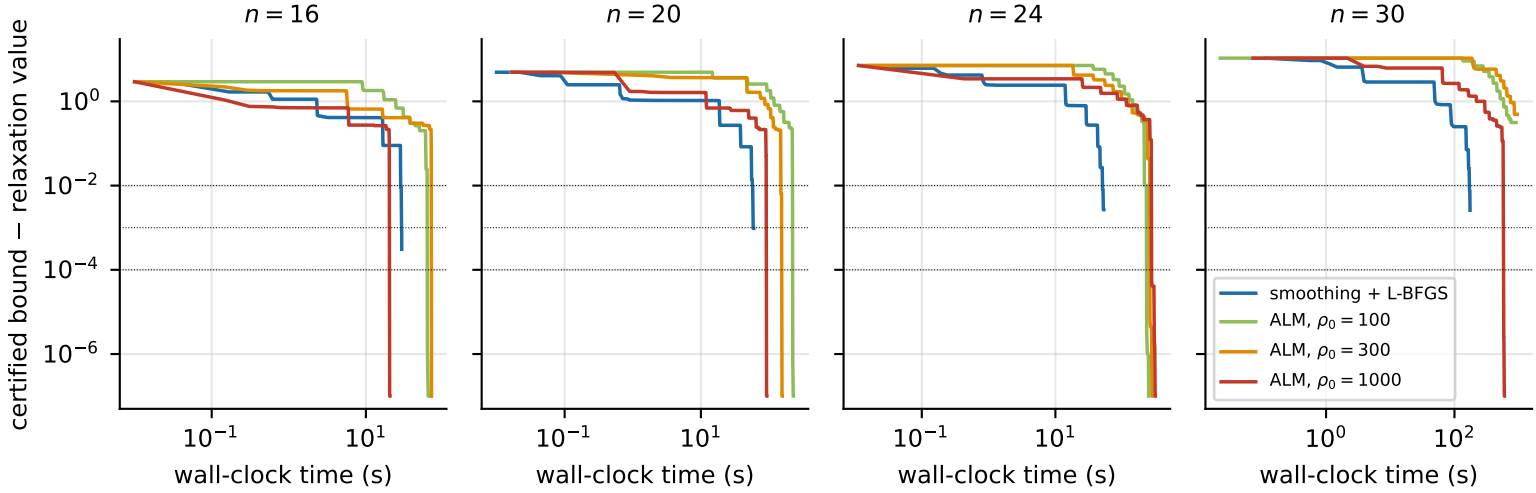
$n = 16$
$n = 20$
$n = 24$
$n = 30$
certified bound − relaxation value
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-1}
10^{1}
10^{0}
10^{2}
wall-clock time (s)
smoothing + L-BFGS
ALM, $\rho_0 = 100$
ALM, $\rho_0 = 300$
ALM, $\rho_0 = 1000$